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Participatory Research Through Design for Societal Change – A Case Study of Co-Designing a Prototype for Enhanced Cognitive Accessibility of an eHealth Service

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ABSTRACT

This study describes the development of a three-step method called Participatory Research through Design using a co-design and dialogue approach to enhance cognitive accessibility in eHealth services in public healthcare. The method is a combination of Participatory Design, Participatory Action Research and Research through Design (RtD). Together with participants with lived experiences of cognitive impairments we explored the existing personal eHealth services with Participatory Cognitive Barrier Walkthrough and co-designed a prototype for enhanced cognitive accessibility. We used an explorative study design constructing the prototype that presents the preferred state, where the participants' contributions are embedded in the design (e.g., visualization of high-cost protection for healthcare, notifications, read aloud functionality, and photos of drugs). The ownership of the targeted eHealth services was outside the research project, and the co-designed prototype was used as a dialogue tool with stakeholders from the product owner perspective. We propose that our Participatory RtD approach enhances understanding of end-user perspectives in a given context and promote dialogue around complex eHealth services.

KEYWORDS

Interface design and evaluation methodologies; human-computer interaction theory; disabled populations; prototyping; user-centered design

1. Introduction

The digital transformation of healthcare is suggested to contribute to achieving the health-related Sustainable Development Goals by the United Nations World Health Organization (2021), such as goal number 3: "Ensure healthy lives and promote well-being for all at all ages" (United Nations, 2015).

However, technology is not neutral and considering who benefits from it is important when addressing sustainability and social justice (Stephanidis et al., 2019). Accessibility barriers in today's increasingly digital world are prevalent (Acosta-Vargas et al., 2024; Alajarmeh, 2022; Paul, 2023; Sarita et al., 2022). Accessibility is acknowledged as key for inclusion according to the Convention on the Rights of Persons with Disabilities (CRPD) (United Nations, 2006). Accessibility barriers are identified as one of the seven Human Computer Interaction (HCI) grand challenges (Stephanidis et al., 2019). Addressing the wicked problem of accessibility, particularly cognitive accessibility in eHealth services, in real-life situations is complex.

Considering that 1.3 billion people in the world experience significant disability (World Health Organization, n.d.) designing accessible eHealth services is vital for all to benefit from the digital transformation in healthcare. There are regulations addressing web accessibility in many parts of the world

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(W3C Web Accessibility Initiative (WAI), n.d.). The European Union (EU) states that digital accessibility is an important aspect in the vision of creating a "Union of equality" (European Commission, 2023).

Accessibility is defined in the Ergonomics standard ISO 26800:2011 as:

> extent to which products, systems, services, environments, and facilities can be used by people from a population with the widest range of characteristics and capabilities to achieve a specified goal in a specified context of use. (International Organization for Standardization, 2011)

The EU regulates a minimum level of accessibility in public sector bodies' websites and mobile applications. The regulation is described in detail in EN 301 459 (European Telecommunications Standards Institute, 2021), which builds on WCAG 2.1 (World Wide Web Consortium (W3C), 2005). However, WCAG 2.1 does not provide sufficient guidance on cognitive accessibility (James et al., 2017; Johansson et al., 2015) and W3C confirms that not all aspects of cognitive accessibility are covered in the existing standards (World Wide Web Consortium (W3C), 2019). People with any impairment have equal rights to use public digital services in the EU. Therefore, the design of public digital services must be accessible from all perspectives, including cognitive accessibility.

The Cognitive Accessibility standard ISO 21801-1:2020 describes cognition as:

> Mental actions or processes of acquiring knowledge and understanding through thought, experience, and the senses.
>
> Note 1 to entry: Specific cognitive functions include perception, attention, memory, linguistic functions (verbal, non-verbal), reasoning, planning, problem solving, decision making, reading and calculating. Cognition interacts with global mental functions (e.g., consciousness, energy, motivation) and affective functions (e.g., emotion, mood and self-regulation). (International Organization for Standardization, 2020)

The Cognitive Accessibility standard (International Organization for Standardization, 2020) takes a holistic view of user experience, focusing on cognitive accessibility but presents abstract information on how to design to achieve cognitive accessibility.

1.1. The (in)accessible design of eHealth applications

Research shows that people with impairments experience more accessibility issues with eHealth compared to people without impairments (Pettersson et al., 2023, 2025). The Web Accessibility Directive in the European Union (EU) regulates mandatory accessibility evaluations of public sector webpages and apps, along with a publicly available Accessibility Statement (European Commission, 2023). The Accessibility Statements should provide information for users about known accessibility issues in existing webpages and apps. A review of Swedish public healthcare providers' Accessibility Statements in Sweden showed that accessibility issues are present in all included eHealth services (Jonsson et al., 2025).

The accessibility issues with existing eHealth may have multiple explanations stemming from design processes. One explanation may be that there is a vast amount of research on eHealth services but limited utilization of accessibility standards and guidelines in the evaluation and development process (Jonsson et al., 2023). Another explanation may stem from the accessibility standards and guidelines. Lewthwaite (2014) explains disability as: "political, culturally contingent, multi-dimensional and highly complex," and argues that the preeminent accessibility standards and guidelines fail to consider the individual's context.

Considering the user in the design process could solve some of this problem. An overview of the concepts relating to different approaches in design is presented here and summarized in Table 1. A literature review by Göttgens and Oertelt-Prigione (2021) shows that healthcare research increasingly uses Human-Centred Design (HCD) (International Organization for Standardization, 2019), User-Centred Design (UCD) (Gulliksen et al., 2003) and Design Thinking (DT). However, van Velsen et al. (2022) discuss that there are several limitations in current HCD and UCD design approaches regarding e.g., bias, risk of supporting the status quo and not involving all stakeholders. The authors suggest design processes could be improved with Value Sensitive Design (VSD), citizen science visioning the future and more-than-human design in eHealth context (van Velsen et al., 2022). The design thinking framework Double Diamond was developed from analyzing how designers work, describing the designers'

Table 1. Overview of concepts relating to different approaches in design.

Design approaches or research methods	Description	Reference
Human-centred design (HCD)	"… Aims to make systems usable and useful by focusing on the users, their needs and requirements, and by applying human factors/ergonomics, and usability knowledge and techniques."	International Organization for Standardization, 2019
User-centred design (UCD)	Design process that focus on users, conducted in different ways both with and without users actively participating in the design process.	Gulliksen & Lantz, 2003
Double Diamond	A design thinking (DT) framework describing the designers' design and innovation process, consisting of four iterative steps: Discover, Define, Develop, and Deliver.	Design Council, n.d.-a. Design Council, n.d.-b.
Value sensitive design (VSD)	An approach to the design of technology that accounts for human values in a principled and comprehensive way throughout the process.	Friedman et al., 2013
Participatory design (PD)	Puts the end-user's needs in focus by bringing them into the research as participants, emphasizing democratic, mutual learning and empowerment.	Bødker et al., 1987 Bødker & Pekkola, 2010 Sundblad, 2011 Bødker et al., 2022
Action research (AR)	"… Methods and approaches for conducting democratic and collaborative research with community partners."	Hayes, 2011
Prototype and prototyping	Prototypes enable design-thinking, provide evolutionary learning, discovery, and stimulate reflection.	Lim et al., 2008
Cooperative prototyping	Users and designers cooperating in activities in the development of the prototype.	Bødker & Grønbæk, 1991
Research through design (RtD)	Scientific inquiry into, through and for design that is both interested in producing knowledge from the design process and by the artifact.	Frayling, 1993 Godin & Zahedi, 2014

design and innovation process (Design Council, n.d.-b.). The Double Diamond consists of four iterative steps: Discover, Define, Develop and Deliver (Design Council, n.d.-a.). The Double Diamond promotes the principle "Be people centred" but it does not alone fully emphasize if the design process is empowering the users or if all users will be able to use the design. This can be exemplified by Barnes and Du Preez (2015) who adapted the Double Diamond focusing on empathy development, moving from what users can bring to the design to how problems can be creatively explored. HCD, UCD and DT may not be enough to ensure designing cognitively accessible eHealth services that empowers the users. Participatory Design (PD) puts the end-user's needs in focus by bringing them into the research as participants, emphasizing democratic, mutual learning and empowerment (Bødker et al., 1987, 2022; Bødker & Pekkola, 2010; Sundblad, 2011). Hayes (2011) says that "the historically related and conceptually intertwined ideals and ideas of Action Research (AR), industrial democracy, and sociotechnical thinking" (2011) contributed to the development of new design methods such as Scandinavian (or Cooperative Design) and Participatory Design (Hayes, 2011). Kurt Lewin is considered to have coined the term "action research" and for the early definitions within the field, including seeing researchers as actively problem solving alongside the "research subjects" (Hayes, 2011). An early example leading the way for PD is the famous UTOPIA project, a research project involving graphic workers at every step of the design and development of the new IT support workstation (Bødker et al., 1987; Sundblad, 2011). The Scandinavian tradition of cooperative or participatory design is a form of user influence in the system development process, rooted in democratizing workplaces (Robertson & Simonsen, 2012). PD has been used in various research disciplines (Bødker & Pekkola, 2010) including healthcare. PD has been used in the development of eHealth interventions, but the reporting of PD methods varies (Vandekerckhove et al., 2020).

Prototypes and prototyping have been widely used in HCI research, significantly contributing to the development of more user-friendly technology (Odom et al., 2016). Prototypes are common in the development of eHealth (Kip et al., 2022). Lim et al. (2008) explain that prototypes in HCI design research enable design-thinking and provide evolutionary learning and discovery. They further state that prototypes stimulate reflection and discovering possibilities. Lim et al. (2008) present a framework with two principles of prototyping and the anatomy of prototypes. The first principle is that prototyping is an activity that filters without losing the understanding of the whole. The second principle targets the economic principle of prototyping, which visualizes possibilities and limitations of a design idea in the simplest and most efficient way and makes it measurable (ibid). Additionally, Lim et al. (2008)

explain the anatomy of prototypes as "filters that traverse a design space and are manifestations of design ideas that concretize and externalize conceptual ideas". Cooperative prototyping makes it possible for mutual learning and can be used as a basis for idea exploration (Bødker & Grønbæk, 1991).

Research through Design (RtD) stems from Frayling who introduced the idea of three kinds of design research: **into, through,** and **for** art and design (Frayling, 1993). Godin and Zahedi (2014) conducted a literature review of RtD research and present a broad field of RtD exposing differences between researchers, but who all still share the goal of "establishing aspects of research done through the design process and its resulting product". The authors explain RtD as a scientific inquiry about design that is both interested in producing knowledge and in the end product (the artifact). Godin and Zahedi (2014) define RtD as "an approach to scientific inquiry that takes advantage of the unique insights gained through design practice to provide a better understanding of complex and future-oriented issues in the design field." RtD hosts possibilities to gain knowledge in the design process of building an artifact and understanding by the artifact, even though the artifact itself is not the goal of RtD (Godin & Zahedi, 2014).

PD, PAR, cooperative prototyping and RtD all allow for design processes that could contribute to designing complex eHealth services that are experienced as cognitively accessible by the user.

1.2. The study context

The digital transformation of healthcare in Sweden, has been ongoing for many years. Public healthcare in Sweden is primarily funded through taxes and available to all residents. Patients pay relatively low fees for healthcare and medication with a maximum cost limit for a 12-month period. Public healthcare is organized by the 21 self-governing regions offering healthcare through both public and private actors. The self-governing regions cooperate through the Swedish Association of Local Authorities and Regions (SALAR). All regions use the national healthcare web portal with open and personal eHealth services available on the webpage 1177.se. The digital ecosystem and the software systems in 1177.se are managed by Inera (n.d.), a company owned by SALAR. All residents in Sweden are potentially affected by the design of the personal eHealth services in 1177.se, for example when accessing their healthcare records or renewing prescriptions for medications.

The ethical approach of public healthcare in Sweden has been shifting to a person-centred care approach, considering a person's context, needs, preferences, and relations (Ekman et al., 2011; Yun & Choi, 2019). Person-centred care is the core of the ongoing transformation to organize and deliver healthcare to a greater extent based on the patient's needs and circumstances (Socialstyrelsen, 2021).

This paper presents a study from the research program CoDeAc, an abbreviation for Co-Design for Accessibility (CoDeAc, n.d.). The program relies on PAR (Macdonald, 2012) and PD (Bødker et al., 2022) and has a democratic approach. CoDeAc is guided by VSD (Friedman et al., 2013) regarding ethical considerations. Participants in the CoDeAc program come from different stakeholder perspectives: user; design; healthcare; technology; and policy/legal perspectives. The CoDeAc program has been ongoing since 2019 and has conducted several studies in different projects about 1177.se and other topics. The projects in CoDeAc have a strong focus on design, visioning better futures with an RtD approach.

This study is grounded in problems brought up by members of the Begripsam Group regarding the Swedish national public healthcare personal eHealth services in 1177.se that were experienced as hard or impossible to use. These experiences are in line with research presented by Johansson et al. (2021) and Pettersson et al. (2023). The personal eHealth services are accessed with electronic identification and consist of a complex digital ecosystem with several different eHealth services e.g., electronic healthcare record and online booking. Public healthcare providers decide which services they offer, which can therefore vary for patients. A brief overview of personal eHealth services in 1177.se is presented in Supplementary Appendix 1, for the reader to better understand the complexity of 1177.se from the patient perspective.

1.3. Aim

This study aims to describe how we developed a method for Participatory RtD which involves a co-design and dialogue approach putting the design in focus for exploring better futures and using design aiming to affect the existing eHealth services in public healthcare. We present a case study of a co-designed prototype with a focus on enhanced cognitive accessibility targeting an already implemented eHealth service in public healthcare.

We describe the end-user perspective mainly from a potential patient's view. However, we do not consider participants in the co-design process as patients, although their lived experiences from healthcare contribute vital knowledge to the co-design process. We consider the participants as co-researchers in the co-design process. The term "patient" in this paper is used to help the reader understand that we focus on the person managing their health.

2. Materials and methods

2.1. Study design

This study falls under the RtD (Frayling, 1993; Redström, 2017; Zimmerman et al., 2007) umbrella to "make the right thing: a product that transforms the world from its current state to a preferred state." (Zimmerman et al., 2007). The RtD approach is combined with PD and AR as the suggested Participatory RtD method. We used an explorative study design constructing an artifact (the prototype) that presents the preferred state, where the participants' contributions are embedded in the design. Data were analyzed by qualitative reflexive thematic analysis in an iterative process using both an deductive and inductive approach following the six steps: (1) familiarizing yourself with your data, (2) generating initial codes, (3) searching for themes, (4) reviewing themes, (5) defining and naming themes, and (6) producing the report by Braun and Clarke (2006). From the insights gained from analyzing the co-design process, the co-designed prototype for enhanced cognitive accessibility and using the prototype as a dialogue tool we formulate a new method that we have called Participatory Research through Design (Participatory RtD).

2.2. Participants

Workshop participants were recruited from Begripsam Group, a member organization working for accessibility and Funktionsrätt Skåne, an umbrella organization for disability organizations in the Skåne region. All members in the Begripsam Group have lived experience of cognitive disability due to different impairments or diagnoses. The participants from Funktionsrätt Skåne have lived experience of cognitive impairment or long-term illness. Five researchers and two designers participated. The participating researchers have diverse backgrounds in HCI, accessibility and medical science. The first and second authors work in public healthcare, one experienced in several of the eHealth services in the investigated digital ecosystem. Of the total number of 41 participants, 26 were women and 15 men, with ages ranging from 20 to 75 years. Thirty-one participants reported lived experience of cognitive impairments relating to language impairments ($n = 12$), intellectual impairments ($n = 2$), attention, executive and memory impairments ($n = 15$), neurological and musculoskeletal impairments ($n = 2$), mental and emotional impairments ($n = 10$), deafness ($n = 1$), and hearing impairment ($n = 1$). Nineteen participants had impairments related to one of these categories, nine participants had a combination of two, and three had a combination of three. Member organization participants' experiences with the personal eHealth services in 1177.se ranged from non-user to frequent user. Participants' experience with the personal eHealth services in 1177.se ranged from non-user use to frequent user. Participants used a broad range of devices (smartphone, tablet or computer), and some individuals shifted between different devices (smartphone and computer).

All participants received written and spoken information about the research study and signed a consent form. Participants in the design team received reimbursement. Ethical approval was granted by the Swedish Ethical Review Authority (grant number 2019–04419).

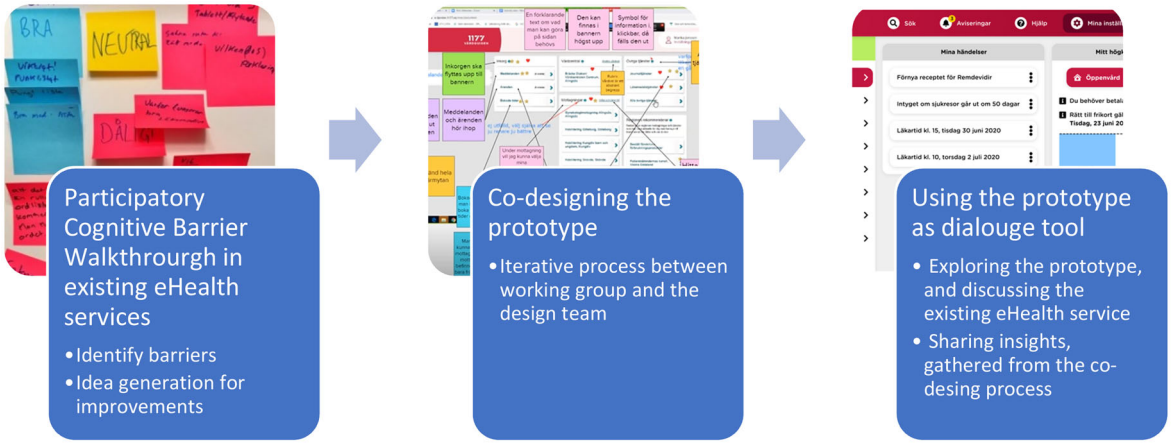


Figure 1. A figure of the process in three steps. The first step explores the existing eHealth service with Participatory cognitive Barrier Walkthrough and generating ideas for improvements. The second step consist of an iterative process between groups co-designing the prototype. The last step consists of using the prototype as a dialogue tool and discussing the existing eHealth services. The last step consisted of both exploring the prototype and sharing insights from the collective knowledge built in the co-design processes.

2.3. Study procedure

The design process was conducted in the following three steps: (I) Participatory Cognitive Barrier Walkthrough (PCBW) in the existing eHealth service, (II) Co-designing the prototype in an iterative process and (III) Using the prototype as a dialogue tool (Figure 1).

In the first step, we investigated participant's experiences of using the personal eHealth services in 1177.se with Participatory Cognitive Barrier Walkthrough (PCBW), which is a method we have developed (Jonsson et al., 2025) from Barrier Walkthrough (Brajnik, 2008) looking at barriers in the current eHealth service. The PCBW, explained in Jonsson et al. (2025), acknowledges the importance of context in line with Brajnik (2008) but we moved to jointly discussing cognitive accessibility barriers during the cognitive barrier walkthrough sessions. PCBW is an evaluation method performed in a group setting together with researchers and people with cognitive impairments, where the participants identify and discuss cognitive accessibility barriers when performing a user scenario in the evaluated system. During the participatory Barrier Walkthroughs participants also generated ideas for improvement. The participatory Barrier Walkthroughs were conducted in groups during a total of 22 workshops, 3 in physical meetings and 19 in online meetings. All groups had persons dedicated to act as workshop leaders and note-takers. The existing eHealth service and the prototype are discussed from a person-centred perspective.

In the second step, we co-designed a prototype for enhanced cognitive accessibility in the personal eHealth services in an iterative process between the working group and a dedicated smaller group, here referenced as the design team. The co-design process was iterative, conducted through workshops that alternated between sessions with the working group and the design team. Attendance at the workshops fluctuated, with approximately 40 participants in the working group and between 4 and 10 participants in the design team. The digital workshops varied in duration, ranging from full-day sessions to brief 1.5-hour meetings.

In the third step, we used the co-designed prototype as a dialogue tool with stakeholders from the product owner perspective to discuss cognitive accessibility in the personal eHealth services in 1177.se. The session was conducted in a Zoom meeting.The workshops and the dialogue session were video-, or audio-recorded and notes were taken during or after the workshops by dedicated participants. The first author revisited the recordings and notes several times during the co-design process and data analysis.

Data analysis was conducted on three separate areas: (1) the co-design process, (2) validation of cognitive accessibility in the prototype, and (3) using the prototype as a dialogue tool.

The research questions explored in data analysis for areas 2 and 3 are related to:

I. what design perspectives do participants with lived experience of cognitive impairment consider enhance cognitive accessibility in eHealth services,
II. the potential of using the co-designed prototype as a dialogue tool with stakeholders.

2.4. Describing the co-design process analysis

Analysis of the co-design process was conducted by the first author, who reviewed all video recordings with a special focus on identifying different forms of design activities and the materials used. A purposeful sample of workshops ($n = 6$) were analyzed using an observational protocol. The observational protocol included date, number of participants, timestamped notes with description of design activity or a description of the ongoing discussion, first authors' notes, and when needed a screenshot of the screenshared content regarding the materials used.

2.5. Validation and analysis of cognitive accessibility in the prototype

To validate that the co-design method used for prototype design targets cognitive accessibility, we used qualitative thematic analysis (Azungah, 2018; Braun & Clarke, 2006) with a deductive approach to coding and interpretation (Braun & Clarke, 2019). The codes were based on the Cognitive Accessibility standard ISO 21801-1:2020 (International Organization for Standardization, 2020). Each item in the standard represented a code. The first author prepared the initial coding, linking one to three features in the prototype to each item in the standard. A screenshot of the specific part of the prototype was posted on a Miro board, and features/design were linked to the number of the item in the Cognitive Accessibility standard. For example, the visualization of high-cost protection and the list of notifications on current and upcoming events on the starting page were linked to item 5.2.1 Recruiting interest (International Organization for Standardization, 2020). All authors reviewed and discussed how the features/design links to the codes until consensus was reached. The design implications for enhancing cognitive accessibility were then grouped into themes using a reflexive thematic analysis (Braun & Clarke, 2019).

2.5.1. Data analysis of using the prototype as a dialogue tool

The first author reviewed the video recording from the meeting with stakeholders from the product owner. The data collection was conducted with an observational protocol looking for tensions, potentials and person-centred perspectives that were presented and discussed in the meeting. Data was collected with time-stamped notes, for example see Table 2 showing a part of the discussion. When needed, screenshots were collected and linked to the time stamped notes, to visualize what was discussed. The data collection was analyzed with a reflexive thematic analysis (Braun & Clarke, 2019).

3. Result

The results are organized in four sections: the co-design process, the co-designed prototype for enhanced cognitive accessibility, using the prototype as a dialogue tool and introducing the three step Participatory Research through Design method.

3.1. The co-design process

The design team started with the design ideas from the working group, developed ideas further and then presented the ideas envisioned in the prototype to the working group. Viewpoints were discussed, and the group collectively decided on the way forward. The process is presented in Figure 2.

Starting with on-site full day workshops in November 2019, we conducted Participatory Cognitive Barrier Walkthrough (PCBW) in the 1177.se, followed by co-designing the prototype. The on-site

Table 2. Example of data collection of a part of one discussion with time-stamped notes.

Time	Transcript	Potential. areas that potentially create solutions, for example potential developmental area based on users' willingness to share data.	Tension area, meaning for example legal, holistic/ person-centred, personal (note the specific personal for example cognitive), system architecture/IT (explain), other	Holistic perspective as in person-centred. For example, holistic perspective of a person and their life, or a holistic view on information and systems such as something needs to interact outside 1177.se (e.g., calendar). Note and explain.
1.54.30	Inera representative: If I try to book that eleven o'clock appointment there, could it be enough that I get a question: "We see that you have an appointment that ends at 8:30. Can you make it on time?" Can that assessment be made? Last author (LA): Yes, it could be a dialogue. We already know you have an appointment. Will you really be able to make it?		Cognitive, time management	Discussing if a question could support individuals regarding time management.
1.54.55	First author (FA): That was something that came up. How is it that healthcare can double-book patients so many times? Shouldn't healthcare know what they are doing?		Patients are double-booked by healthcare.	
	The patient is unaware that healthcare providers are not allowed to access information from other healthcare entities.		Information sharing regulation	
1.55.29	Last author: People would agree to share if asked.	User consent to sharing data		

The discussion is divided into meaning units in each row of the table. Notes representing the codes are made in the relevant column.

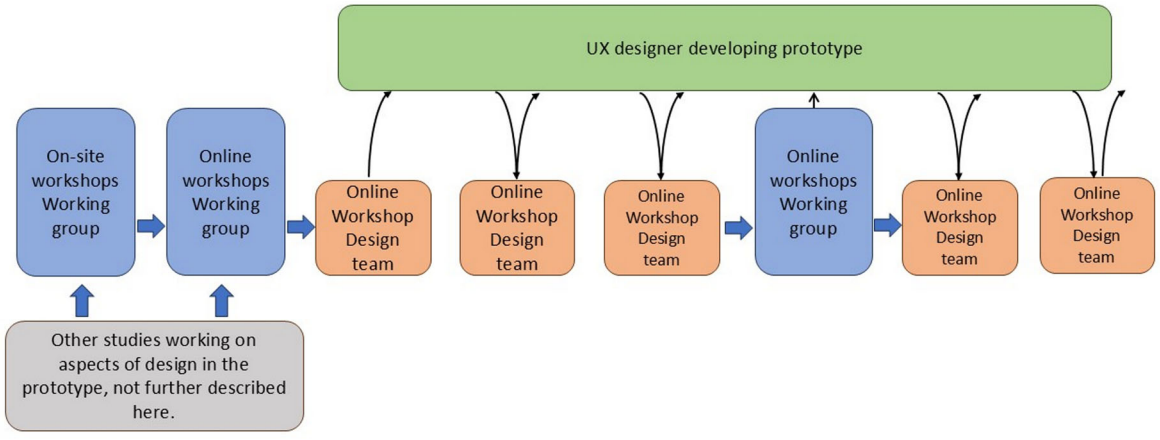


Figure 2. A schematic presentation of the iterative process developing the 1177 prototype. Other studies informed the design of the prototype. The design process shifted between the working group and the design team. The prototype was built by a UX designer and the last author who is an accessibility expert between the sessions.

meetings were carefully scaffolded following descriptions by Johansson et al. (2019). In March 2020, the COVID-19 pandemic forced all workshops to shift to online meetings to preserve physical distance. The transition to online meetings was a joint effort by participants that included adapting scaffolding techniques for online collaboration, handling accessibility issues, providing instructions and educating participants on how to use Zoom and online collaboration tools (Johansson et al., 2025). CoDeAc provided technology (e.g., headphones) and covered the cost of internet access for participants when needed. Initially, we used Mural as the collaborative tool for co-design but later switched to Miro

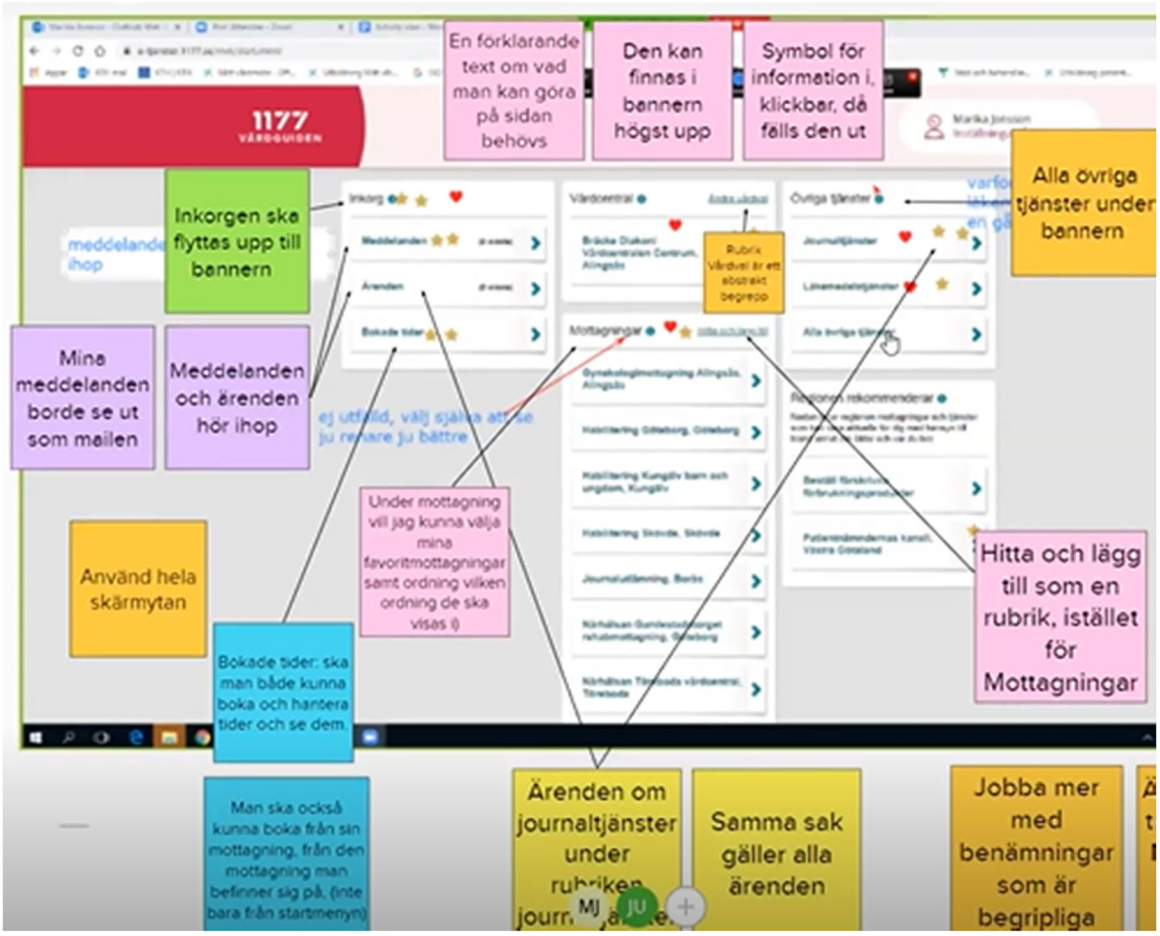


Figure 3. An example of notes on the screenshared view, created during the PCBW of the existing personal eHealth services in 1177.se. Notes were written on digital sticky notes, when needed the sticky note is linked to a specific menu option. The hearts and stars were used for voting.

(Miro, n.d.). Some participants were also involved in other design projects. This allowed them to also bring experiences and reflections from those projects.

All workshops were prepared by the workshop leaders and the UX designer discussing the agenda and preparations. The workshop leader or the UX designer prepared a board in Mural or Miro. As one example preparations consisted of a frame for the design and empty sticky notes, pre-named elements and empty elements that participants could edit and move into the frame.

Each workshop began with brief background information or a review of the previous meeting(s) to ensure we remembered where we left off. The review could consist of demonstrating the previous design and summarizing discussions.

After the initial working group workshops with PCBW a decision was made to co-design a prototype focusing on the starting page, booking appointments, and renewing drug prescriptions. The idea was that these could serve as knowledge base or building blocks to design other services.

Following the iterative process between the working group and the design team, the design team presented the first version of the prototype in the summer of 2020. The working group reviewed the prototype, resulting in a decision to make major redesigns of the drug renewal process. The design team continued redesigning and developing the prototype, finalized by workshops in the working group.

Co-design sessions were conducted in the following ways:

- A PCBW combined with co-design consisting of adding notes and sketching, in physical meetings, or on digital whiteboards and on the screen-shared view (example in Figure 3).
- Simultaneous co-design activities in the collaborative online tool.

- Co-design activities with a dedicated person/persons creating the design from instructions and discussions from all participants during screen-sharing the design activity.
- A flow model created by the UX designer was used in the design team workshops to support the major re-design of the drug renewal process.
- Discussing and choosing design elements pre-prepared by the UX- designer.
- Evaluating and discussing the prototype regarding design decisions and further development.

The prototype and the process map were built in Adobe XD, a tool for designing interactive prototypes, by a UX designer and the last author between the workshops. Design-decisions were discussed and when needed voting was arranged in the online collaboration tool.

3.2. The co-designed prototype for enhanced cognitive accessibility

The co-designed prototype is designed to enhance cognitive accessibility, presented in a high-fidelity clickable prototype in Adobe XD. The prototype is intended for computer-use, presenting the co-designed eHealth services chosen by participants. Parts of the prototype were not fully developed during the co-design process.

Design implications for enhancing cognitive accessibility in the personal eHealth services refer to Personal finance, Processing information, Organization and planning, Support, and Re-using personal data, shown in Figure 4.

3.2.1. Personal finance

The starting page shows the high-cost protection for healthcare visits (Figure 5(D–F)), reflecting the importance of healthcare costs affecting personal finances. The importance of transparent and understandable healthcare costs is shown in several places in the prototype such as the price for a consultation, the renewal of drug prescriptions, and the price of the drug.

3.2.2. Processing information

Images and symbols to support information processing were generally considered helpful. The innovative approach of incorporating representational photos of the drug into the prototype, both of the package and the pill (or other format) was deemed important, supporting both the drug renewal process and daily life.

The first symbol used in the prototype for medication consisted of two black and white pills. One participant in the working group thought the symbol looked like a pair of slippers. The symbol was replaced with a bottle marked with a cross and a pill beside it (see Figure 5(B)).

Participants co-created a familiar design with a logic they could navigate. The starting page mirrors the logic of an email inbox on a web browser. Personalization was suggested to address personal requirements. Therefore, it was proposed that features on the starting page could be personalized by choice (Figure 5(A)).

Understanding and processing time were considered challenging for many. The prototype visualizes time both regarding the time of day and the date in the process of booking appointments. The time of day is visualized with both digital time and an analogue clock. The date is visualized with a calendar with colors and symbols. The weekdays have different colors following the recommendations by The National Agency for Special Needs Education and Schools (n.d.). Available, occupied and chosen dates are coded with color and symbols. Explanations for the color code and symbols are visible in the prototype. Today is marked with a frame around the date. Participants expressed the need for personalization, acknowledging that people may have different preferences regarding color and how time should be displayed (analogue or digital).

Participants reflected on digitalization in society and suggested that all public service providers in Sweden should use the same terms, logic and structure in digital services for similar activities. Participants recalled that the term for messages within different services were for example mailbox, inbox, messages, or my messages. Using the same terms, logic, and structure would make it easier to recognize and use the functionality in different public sector services.



Figure 4. A process chart showing important aspects to achieve and maintain health supported by the personal eHealth services 1177.se including support, personal finance, processing information, organization and planning assistance and re-using personal data with examples.

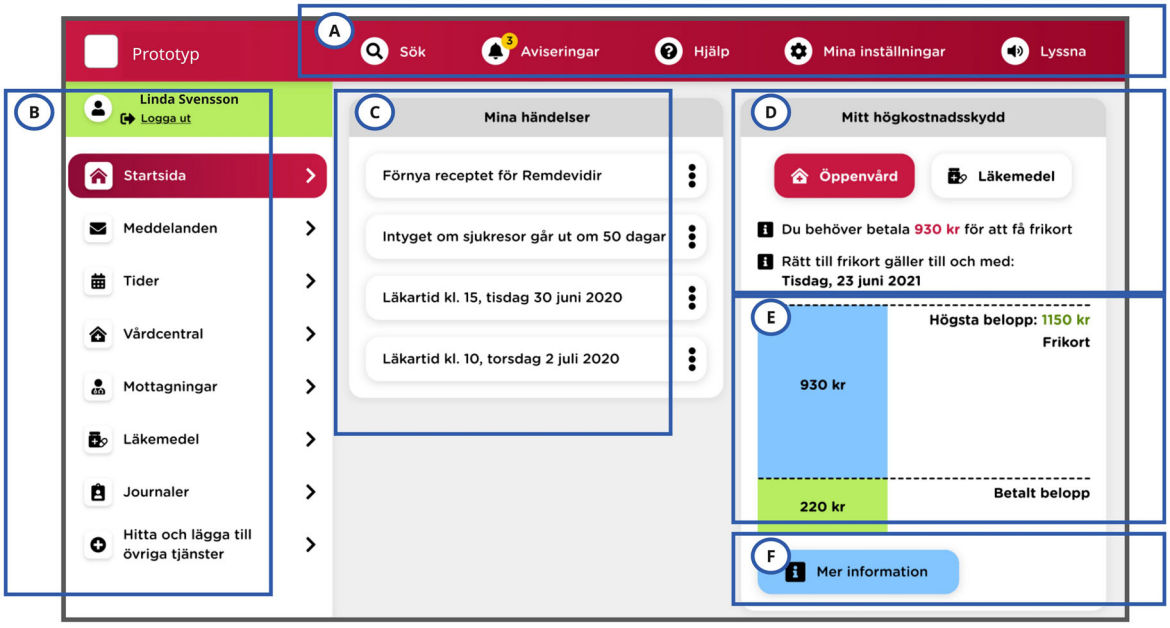


Figure 5. Starting page in the 1177 prototype, design mirroring an email box on a computer. Translations or explanations for each section: (A) search bar, notifications, support, personal settings and read aloud functionality. (B) Name, log out, starting page, messages, scheduled bookings, primary health care clinic, clinics, medication, healthcare record, find and add other services, (C) my events, with a list of reminders in chronological order, (D) outpatient high-cost protection is chosen, information provided regarding the user need to pay 930 SKr to get high-cost protection that is then valid until tuesday 23 june 2021 (E) visualization of high-cost protection for outpatient healthcare (F) more information about the high-cost protection.

3.2.3. Organization and planning

The system should provide notifications on current and upcoming events, along with alerts for actions such as renewing a prescription for a drug that is running low (Figure 5(C)). The system should calculate when the drug is about to run out and present alerts, but the user should also be able to insert alerts by choice. The modality for notification should be chosen by the user, with options including SMS, email, and notifications in the prototype.

Participants emphasized the importance of being able to choose whether the booking should be as soon as possible or with a specific person. Depending on the issue, it could be more important to meet the specific healthcare professional who treated them before for knowledge of medical history or prioritize a consultation as soon as possible with a healthcare professional.

3.2.4. Support

Participants identified and discussed various perspectives regarding support (Figure 5(A)), emphasizing the importance of personal choices in how to receive assistance. The need for support could be technical or healthcare related. Participants highlighted the significance of having representatives, such as legal representatives, guardians, or self-chosen individuals, to support all inquiries, while being able to limit access to specific parts of the service. Participants suggested possibilities for direct calls or chats through the prototype for tech professionals for technical support and healthcare professionals for health-related questions, such as assistance with self-assessment questionnaires. Additionally, participants suggested AI assistance through chatbots and automatic detection of user difficulties, providing help from a person or AI.

3.2.5. Re-using personal data

The process for renewing drug prescriptions was designed to help the user at every step. Information about the drug and where it was prescribed should be retrieved from the healthcare records. A doctor may want to know if the patient experienced side effects. Participants mentioned difficulties in remembering and identifying side-effects. Therefore, options for common side-effects of the specific drug were

presented. This feature was decided to be a personalized choice that can be hidden, if the user does not want information about documented side-effects for the drug. An option with free text is also available, along with speech-to-text functionality and read aloud features. The read aloud feature works for both all text visible in the system and user-generated text.

3.3. Using the prototype as a dialogue tool

The discussions around the prototype provided opportunities to talk about a wide range of areas. The prototype was presented as an example of what is important to enhance cognitive accessibility in personal eHealth services, not as the only solution for how the design should look. Information was given that the prototype was incomplete and contained unresolved design issues. Generally, discussions started with the prototype and led to reflections about the existing system.

Three themes were constructed from the identified tensions: Understandability, Economy and Regulatory perspectives.

3.3.1. Understandability

The theme understandability related to unclear information, time being a complex phenomenon, the system's limitations, and mental models.

Unclear information can arise from differences in how healthcare providers at regional and local levels use and present information in the existing eHealth services, causing trouble for patients due to inconsistency. Dialogue participants reflected on the regions being self-governing and recognized this as part of the problem. Unclear information can also relate to understanding time (e.g., counting backwards due to instructions such as stopping medication X hours before the appointment).

The complexity of the concept of time was discussed when exploring the booking appointment in the prototype. Not all people understand digital time, so the prototype included an analogue clock. The last author presented this part of the prototype as not fully developed due to time being a complex problem that needed further exploration. The co-design participants wanted to know how long an appointment would be. The dialogue participants discussed that the patient and the healthcare professional may not have the same understanding of the length of time for a visit. For a healthcare professional the appointment may include both preparation and post-visit documentation, whereas for the patient, time is related to the actual visit. Dialogue participants talked about difficulties in providing the patient with the actual time for the visit, referring to the system's limitations, considering that healthcare professionals' calendars need to account for preparation and post-visit documentation. Limitations due to information exchange between different IT systems (e.g., healthcare professionals' calendars can be in different systems depending on the healthcare provider) where discussed.

The existing eHealth system's limitations that hinder enhanced cognitive accessibility, such as picture support due to the non-existing image modality was discussed. Mental models related to system developers and healthcare may lack awareness of the patient's situation, and the patient may lack knowledge about healthcare's organization and processes. One example of a knowledge gap is that one of the dialogue participants was unaware that all residents in Sweden, including those who are homeless, must register an address. The existing eHealth service does not provide information indicating a person's homelessness. The last author discussed issues described by homeless individuals in an earlier research project. When the healthcare provider sends an appointment letter, it is likely that the letter will not be received by the patient, resulting in a no-show and the healthcare provider subsequently sending a fee for the missed appointment to the patient. The first author discussed the need for functionality that allows patients to confirm their appointments, ensuring that the appointment is made in agreement between the healthcare provider and the patient.

3.3.2. Economy

The tensions discussed in the theme economy related to healthcare routines and processes causing financial consequences for patients, economy constrains healthcare, and economy constrains the development of 1177.se.

Healthcare routines and processes causing financial consequences for patients due to no-shows being charged with a fee, fees can be sent even if the appointment was made without agreement. Economy

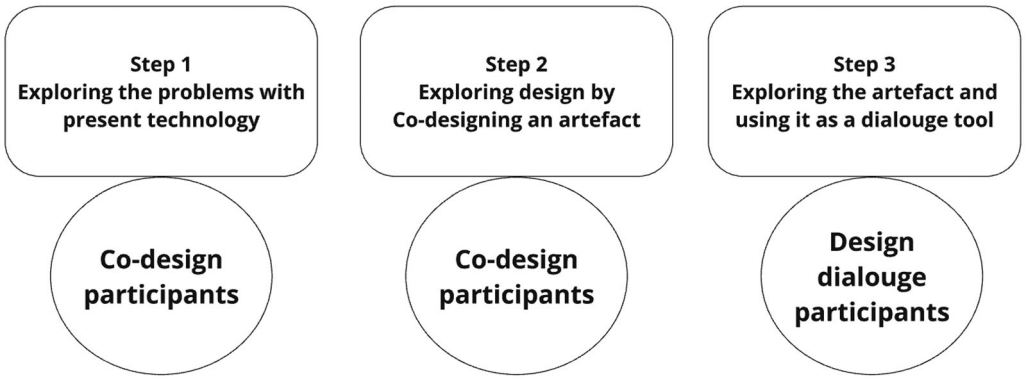


Figure 6. Three-step model for the Participatory RtD process, starting with exploring barriers in existing technology, moving on to co-designing an artifact, and in the last step exploring the artifact using it as a dialogue tool. The first and second steps are conducted with co-design participants. The third step is conducted with design dialogue participants e.g., stakeholders from the product owner perspective.

constrains healthcare was discussed in the difference in fee for video consultations and no fee for phone appointments could motivate healthcare to book video meetings instead of phone calls. This has a financial consequence for patients. The patient's finances could cause the patient not affording getting prescribed medication before the medication runs out. Economy constrains the development of the existing eHealth services, so there is a need to prioritize what to develop.

3.3.3. Regulatory perspectives

Tensions regarding legal perspectives related to gaps in regulation and regulation creating hinders. Current regulation does not sufficiently guide cognitive accessibility. Discussed regulatory barriers include prohibition of sharing prescribed medication information within the same system, and the strict regulations on information exchange between systems, authorities and healthcare in Sweden. The information exchange is often prohibited even if the patient would want it and willingly consent. The regulation requiring a homeless person to have an address, does not reflect their real-life situation.

3.4. Participatory research through design

As shown, we combined PD, PAR and RtD in our study. From that we propose a method in three steps, described in the model in Figure 6. The method explored design and had focus on design at every step. The participants in the different steps may vary and have different roles. The explored research questions can differ in the last step depending on the participating representatives, as in what perspective the participants have. The participants in the first two steps are named co-design participants and in the last step Design dialogue participants.

In Participatory RtD, the explored problems in technology should target real life issues and be investigated together with people who are affected by the problems. When engaging in co-design activities, the participants should be involved in a democratic manner, following the framework for co-designing with extreme users (Johansson et al., 2023). When engaging in exploring the artifact together with Design dialogue participants, the researcher's role should focus on bringing the co-designed participants' voices into the room by facilitating discussions starting from the artifact. The discussions should aim for shared knowledge, finding tension areas, and possibilities to change the present state (in this study, the existing personal eHealth services in 1177.se). The researchers should provide storytelling around the artifact to broaden and enhance understanding around issues and possible solutions.

4. Discussion

This paper contributes in three ways to the research field of HCI and interaction design: (i) Describing a new method, Participatory RtD, aimed at affecting existing eHealth services by using the co-designed prototype as a dialogue tool, (ii) Presenting a case study for on-site and remote participatory co-design

of complex eHealth services, and (iii) Providing design implications for enhanced cognitive accessibility in complex personal eHealth services.

4.1. Participatory research through design

The three-step method, combining PD, PAR, and RtD, addresses real-life problems and aims for visioning better futures to make a real impact. We propose calling this method Participatory RtD, suitable for addressing already implemented complex eHealth services where the ownership of the eHealth service is outside the research project. This method involves exploring existing services and co-designed prototypes with both co-designers and external participants in different settings with design in focus. This is in line with how Godin and Zahedi (2014) describe RtD as a scientific inquiry focused on design, producing knowledge through the design process, with interest in the artifact still not viewing the artifact as the primary goal.

Distribution of power is important in the introduced Participatory RtD method, enhanced through the dialogue with stakeholders from the product owner perspective via the prototype. Bratteteig and Wagner (2012) discuss the sharing of power in PD and say that the artifact itself has a voice and may be a part of the decision-making. Our study had a somewhat different approach, when using our artifact to affect another artifact (the existing personal eHealth services in 1177.se). Bødker and Kyng (2018) argued that PD projects should place the user's voice in a context of potential conflict. We used the co-designed prototype as a dialogue tool with stakeholders from the product owner perspective, bringing the user's voice embedded in the artifact to the context of potential conflict. We consider these conversations as part of what designing is, bringing views and design decisions of the people who are (potential) users of the existing system to the stakeholders who have the power to change the design in the existing system. We consider this approach as a way of making the voice heard from the people not (yet) invited in the design-decision process and a potential way of distributing power.

We do not see the prototype as a stand-alone product when exploring it with dialogue participants, instead it serves as a dialogue tool to create opportunities for discussion. The prototype visualizes design improvements from the end-user perspective, enhancing cognitive accessibility and usability. However, the prototype does not cover all aspects discussed in workshops. When using the prototype a dialogue tool, we also discussed features not included in the prototype that were brought up in the co-design process, such as personalization and support options. The prototype serves as a changing artifact, it can and has been developed further after the case study presented here. In the re-invented PD that matters Bødker and Kyng (2018) argue that working prototypes should be deployed. Although we constructed a prototype that wasn't deployed in the real system, we argue that our proposed way of using a prototype as a dialogue tool can make change in the real system by spreading knowledge and promoting cognitive accessibility.

Exploring the existing personal eHealth services and the prototype helped build a shared understanding during the co-design workshops. The use of a flow model also helped establish a common understanding of complex processes among co-design participants. Building on Lim et al. (2008) framework, we used the prototype for evaluation, understanding user experience, idea generation, and communication among all stakeholders even though all stakeholders did not meet each other. The prototype was a powerful tool for dialogue, sharing insights about solving lived experiences of disability and cognitive accessibility barriers. The prototype served as a common ground and a vision of a better future. It enabled respectful sharing of stories from the collective knowledge without exposing individual participants. This aligns with Gulliksen and Lantz (2003) suggestion that prototypes help build a shared understanding by providing concrete features for discussion, rather than offering the only solution to a problem. Using the prototype as a dialogue tool with stakeholders from the product owner perspective provided possibilities for mutual learning and knowledge building about cognitive accessibility in personal eHealth services. However, we do not know the extent of the dialogue's impact.

The presented Participatory RtD method provide the opportunity to engage in the evaluation and development of large-scale systems, already existing in society. We propose the Participatory RtD method could be used for other large-scale systems than eHealth. Our method can be used for research

with both participants actively involved in the design process and participants reflecting on the co-designed artifact.

4.2. The design process

In PD distribution of power in design-making processes is important (Bratteteig & Wagner, 2012). The methods used for co-designing the prototype facilitated a distribution of power during the co-design. Our starting point was that all participants were considered equal co-researchers. During the whole process, power relations were constantly reflected on, for example in how all participants had a voice in the process along with practical adaptations about how to request or be given the floor. The prototype was developed through a series of workshops and iterative design processes between the design team and the working group. Although the design team focused on designing, power was distributed in the working group discussions. Critique from the working group workshops led to further development of the prototype, particularly in the process for renewing drug prescriptions. The design team reflected on why the booking appointments design in the prototype was successful but not the renewing drug prescriptions. Both had started with the PCBW method. The discussions revealed higher complexity in renewing drug prescriptions compared to booking appointments and a lack of shared understanding or mental model (Carroll & Olson, 1988, p. 47) of the process. Some participants had stated that they failed to acquire the drug in the existing eHealth service. The designer reflected that they were not familiar with the process of renewing drug prescriptions and experienced that it was harder to understand the participants' design requirements for renewing drug prescriptions compared to booking appointments. From the reflections, the design team suggested that a flow model is needed for complex processes to facilitate shared understanding and developing the individuals' mental model of the process but deemed that this step was unnecessary when designing well-known simpler processes where the participants already shared understanding. The flow model step can, as in our case, be used to solve design problems that appear when developing the prototype. However, some participants discussed the flow model, reflecting that it was harder to understand and wondered for example what the different colors meant. The design team decided that before working with a flow model, it needs to be properly introduced. More research is needed to determine beforehand when the flow model is needed to avoid design setbacks. Exploring the existing personal eHealth services and the prototype helped build a shared understanding during the co-design session with co-design participants. The use of a flow model also helped establish a common understanding of complex processes. In relation to Lim et al. (2008) framework, we used the prototype for evaluation, understanding user experience, idea generation, and communication among all stakeholders even though all stakeholders did not meet each other. The prototype was a powerful tool for dialogue, sharing insights about solving lived experiences of disability and cognitive accessibility barriers. The prototype served as a common ground and a vision of a better future. It enabled respectful sharing insights from the collective knowledge without exposing individual participants.

5. Accessibility in eHealth

Our findings on accessibility issues in the personal eHealth services in 1177.se are in line with previous research showing usability issues reported by users for a specific eHealth service in 1177.se, the healthcare record (Hägglund & Scandurra, 2022). Usability and accessibility overlap but are not the same. Hägglund and Scandurra (2022) concluded that data collected by using the System Usability Scale (SUS) did not give enough information about what types of usability problems end users experience or the severity of a problem. Hence it is a need to go deeper, especially to understand usability problems experienced by users with impairments. Petrie and Bevan (2009) compared the Web Accessibility Initiative (WAI) definition of accessibility with the definition by ISO 9241 – 171 and explain that depending on the definition of accessibility, usability can be seen as a sub-set of accessibility as in the case with the definition by ISO 9241-171, or the other way around as in the case with the WAI definition (Petrie & Bevan, 2009). One important aspect is that accessibility is regulated by law in the EU and in Sweden, whereas usability is not. Our study focused on the starting page, booking appointments, and renewing drug prescriptions and did not explore the healthcare record. However, the healthcare

record is part of the investigated digital ecosystem in 1177.se and therefore it is reasonable to argue that the findings by Hägglund and Scandurra (2022) support our findings. Hägglund and Scandurra (2022) conclude that to ensure all patients benefit from healthcare record access, it's crucial to understand the usability challenges and how policies, regulations, and technical decisions impact the effectiveness of record access. Our study confirms that cognitive accessibility issues cannot be solved by design alone. Changes in healthcare work processes and regulations may be necessary. We also argue that it is important to understand accessibility, specifically cognitive accessibility challenges to ensure all patients benefit from the personal eHealth services in 1177.se

There is limited information and education for residents/patients provided by Inera regarding how personal services in 1177.se function or how public healthcare are organized in Sweden. There may be information and/or education regarding a specific eHealth service or a specific part of the service, but no overview or onboarding for users to understand the personal eHealth services in 1177.se from a socio-technical perspective. We suggest that there is a lack of comprehensive information and education for residents on how personal eHealth services in 1177.se function and how public healthcare is organized in Sweden. Hägglund and Scandurra (2022) also suggested that patient education may be needed for the healthcare record.

Residents in Sweden often rely on the personal eHealth services in 1177.se, with limited alternatives for digital communication with public healthcare. Access to healthcare through telephone may also be restricted. Therefore, it is vital that eHealth services can be used by the widest range of users. Our study revealed many barriers relating to cognitive accessibility. Hägglund et al. (2022) argue that online access to health records is a key for empowering patients in their healthcare. Santana et al. (2018) proposed the creation of a common eHealth platform as a key element in the framework for person-centred care. Our study revealed that current personal eHealth services do not fully support person-centred care. The personal e-Health services in 1177.se do not consider the person's context or re-use personal information and the system does not aid or support personal needs or preferences, nor relations in the care and we argue that designing the common eHealth platform should be done considering cognitive accessibility and involving people with lived experience of disability due to cognitive impairments. For example, healthcare fees affecting personal finances were considered an important part of the personal eHealth services and is one of the suggested features that consider the person's context and support personal needs or preferences. Therefore, we argue that end-users with cognitive impairments should be actively involved in developing and evaluating personal eHealth services, for both enhanced experiences of cognitive accessibility and for enhanced support of person-centred care.

6. Strengths and limitations

Participants from the Begripsam Group and Funktionsrätt Skåne have lived experiences with a wide range of impairments/diagnosis. Some participants could not access the personal eHealth services in 1177.se due being excluded from having a digital identification. We consider this a strength, bringing multiple experiences of the eHealth service into the co-design process. When designing a complex service that targets potentially all people in Sweden, we believe that designing for people with most difficulties will help all users. We also consider the broad and deep knowledge in the researcher's group as a strength. Personal eHealth services in 1177.se are complex relating to a wide range of perspectives both regarding technical, organizational and relating to the digital eco system.

The PCBW method generated rich data by simultaneously discussing issues and co-designing during the walkthrough. This approach reduced cognitive load related to memory retrieval and allowed participants to reflect on their previous interactions with the service.

One limitation is that we have not considered the healthcare professional's interface or perspective of the personal eHealth services in 1177.se. This study has limitations regarding that we did not explore other languages than Swedish. More research is needed in these areas.

The prototype itself has limitations. For example, we did not co-design the specific symbols and images but use these to communicate the importance of symbols and images. Choosing from a set of existing symbols sets boundaries to the imagined future. However, we gained important knowledge about how symbols could be misinterpreted and therefore, we suggest further research in this area.

7. Future research

Future research should evaluate the three-step method in other cases. Future research should also investigate the impact related to using the prototype as a dialogue tool. Future research is needed to explore how educational resources can be designed to support users' understanding of the digital ecosystem in 1177.se, and how to design and implement support systems to enhance cognitive accessibility in eHealth services.

Additionally, we suggest that changes made in the personal eHealth services in 1177.se should be evaluated regarding both usability and accessibility.

8. Conclusion

Our Participatory RtD method provides opportunities to engage in the evaluation, development and redesign of already implemented publicly available eHealth services, where the ownership is outside the research project. By involving end-users and stakeholders from the product owner perspective, we aim to create a more inclusive and accessible eHealth service, democratizing the design process. In a world of fast evolving technology and a changing society, we propose that our Participatory RtD approach can contribute to the understanding of end-user perspectives in a given context and create possibilities for dialogue and understanding of the complex phenomena of personal eHealth services.

To achieve the United Nations Sustainability Development Goal number 3 (United Nations, 2015), we propose that eHealth services should be evaluated and designed together with people with lived experience of cognitive impairment. Merely complying with the WAD (European Commission, 2023) on minimum level is not enough to secure cognitive accessibility.

The personal eHealth services in 1177.se we explored did not fully support person-centred care. Designing for enhanced cognitive accessibility involving participants with cognitive impairment is a vital part to succeed in this matter. Regulatory changes in Sweden will be needed to realize some of the suggested features that enhance cognitive accessibility.

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Ethical approval

Ethical approval was granted by Swedish Ethical Review Authority reference number 2019–04419.

Author contributions

CRediT: **Marika Jonsson**: Conceptualization, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Validation, Visualization, Writing – original draft, Writing – review & editing; **Catharina Gustavsson**: Conceptualization, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing; **Jan Gulliksen**: Conceptualization, Formal analysis, Funding acquisition, Investigation, Methodology, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing; **Stefan Johansson**: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Resources, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing.

Disclosure statement

First and second author work in the public healthcare sector, Region Västra Götaland and Region Dalarna. First author worked as a Developmental Manager with implementation and development of eHealth and is now working as Accessibility Strategist. The employers' organizations had no role in the undertaking of this study.

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